(Segment-2) Information Gathering

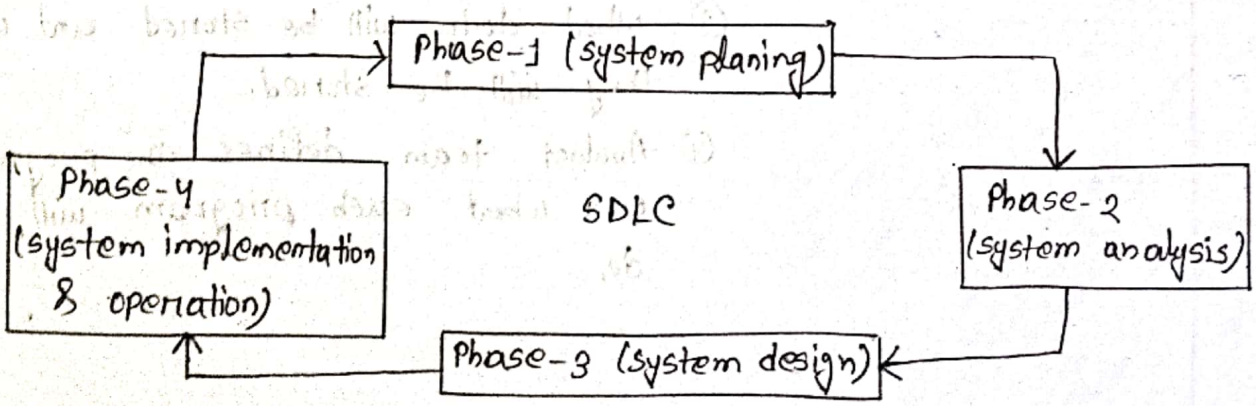
1. Software engineering process
2. SDLC - 4 phases
3. Predictive sdlc vs adaptive sdlc
4. Development methodologies
5. Waterfall model
6. Prototype
7. RAD, JAD, case tools
8. Agile method
9. Agile vs waterfall
10. Spiral model
11. Model, Tools, Technique
12. Increment, Iterative development

Software engineering process: A software engineering process is the model chosen for managing the creation of software from initial customer inception to the release of the finished product. Involves techniques are,

- Analysis
- Coding
- Maintenance
- Design
- Testing

Software development life cycle (SDLC): The SDLC is the process of determining how an information system (IS) can support business needs, designing the system building it and delivering it to users.

4 phases of SDLC,



Phase 1 (Planning): It is the fundamental process of understanding why an information system should be built and determining how the project team will go about building it.

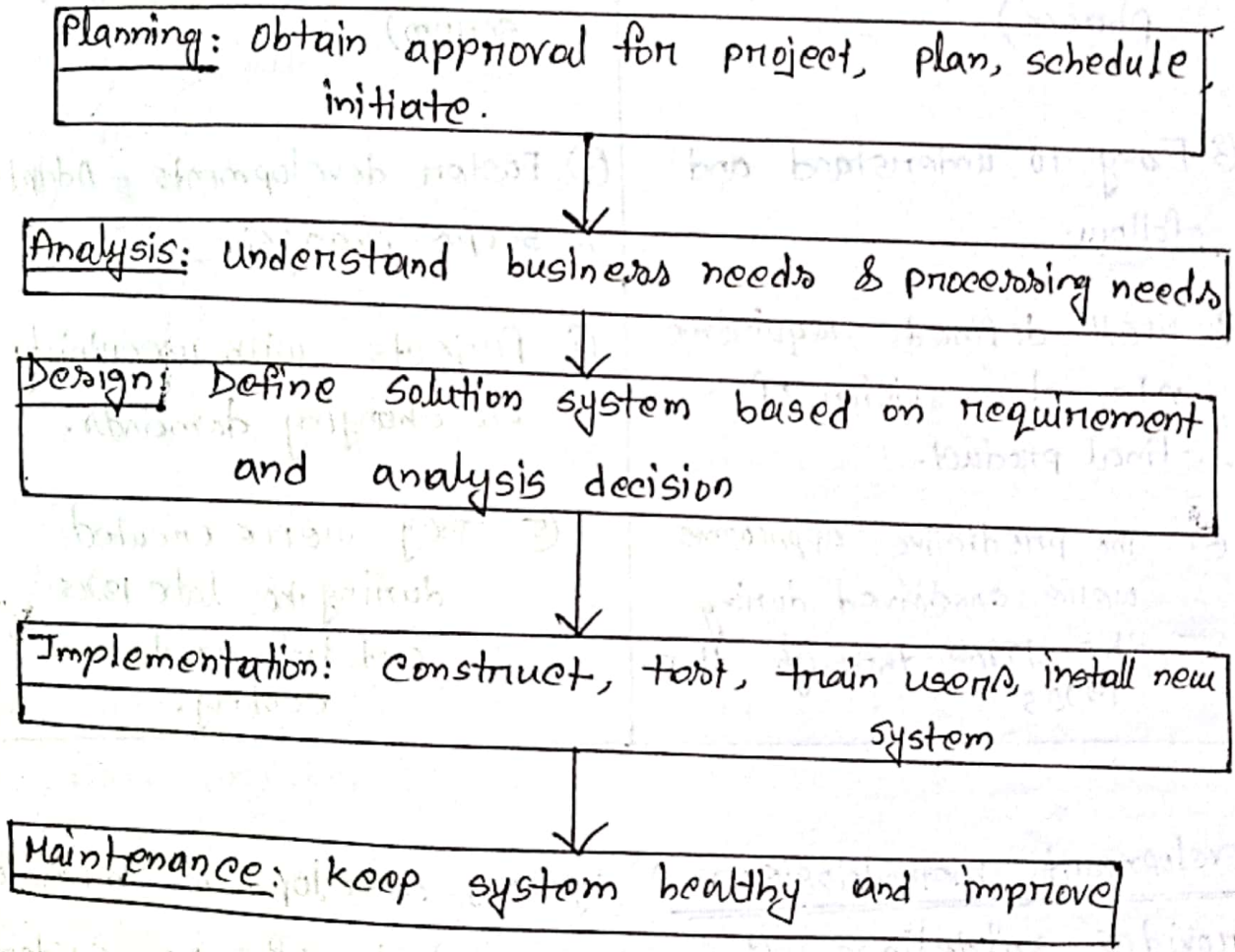
Feasibility analysis – ① Technical feasibility (can we build it?)
② Economic feasibility (will it provide business value?)
③ Organizational feasibility (If we build it, will it be used?)

Phase 2 (system analysis): The analysis phase answers the question who will use the system, what the system will do and where and when it will be used.

The phase has 3 steps: ① study of the current system (as-is system), design a new system (to be system)
② Requirement gathering
③ system proposal which is presented to the project sponsor.

Phase 3 (System design): The design phase decides how the system will operate in terms of hardware, software. 4 steps: ① Must be determined.
② Basic architecture of the design
③ what data will be stored and where they will be stored.
④ Analyst team defines the program and what each program will do.

Phase 4 (system implementation and operation): The final phase is 2 step process. ① Implementation → coding, testing, installation, implementati
② operation



Predictive SDLC Vs Adaptive SDLC:

Predictive sdlc	Adaptive sdlc
① Assumes complete workflow Predictability.	① Adapts to changing needs and uncertainties.
② waterfall model (sequential phases)	② Iterative models (Agile, Scrum)
③ Easy to understand and follow.	③ Faster developments - Adapts to scope changes.
④ well defined requirements, clear vision of final product.	④ Projects with uncertainty on changing demands.
⑤ The predictive approaches were conceived during the 1970s through the 1990s	⑤ They were created during the late 1990s and into 21st century.

Development methodologies: A system development methodology provides guidelines for every fact of the systems development life cycle. It provides a set of comprehensive guidelines for the SDLC that includes specific models, tools and techniques

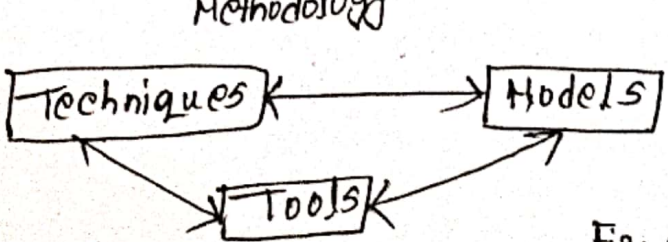
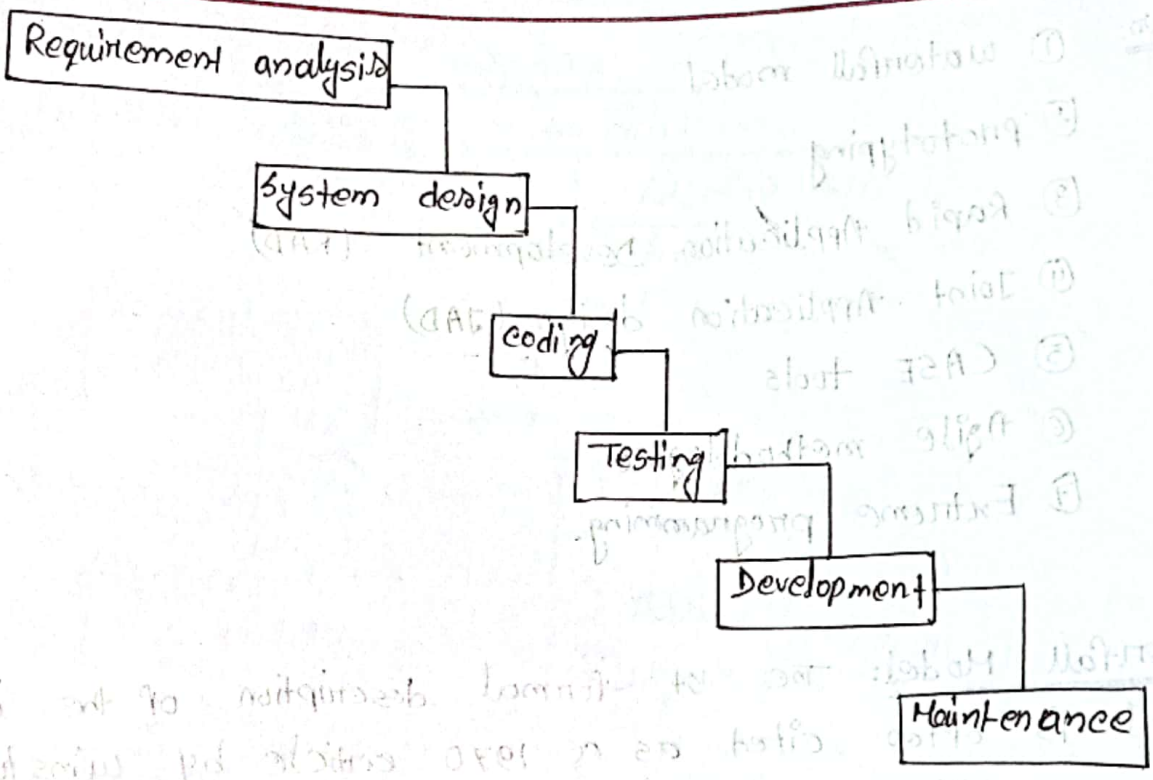


Fig: Components of methodology.

Types: ① Waterfall model
② Prototyping
③ Rapid Application Development (RAD)
④ Joint Application design (JAD)
⑤ CASE tools
⑥ Agile methodologies
⑦ Extreme programming.

Waterfall Model: The 1st formal description of the waterfall model is often cited as a 1970 article by Winston W. Royce. The waterfall model illustrates the software development process in a linear sequential flow; it is also referred to as a linear-sequential life cycle model.
This type of model is basically used for the project which is small and there are no uncertaint requirements
⇒ It was the 1st SDLC model to be ~~used~~ used widely in software engineering to ensure success of the project.
In this approach, the whole process of software development is divided into separate phases.

System and software requirements: captured in a product requirements document.
Analysis: Resulting in models, schema, & business rules
Design: Resulting in the software architecture
Coding: The development, proving and integration of software
Testing: systemic discovery, debugging of defects
Operations: The installation, migration, support and maintenance of complete system.

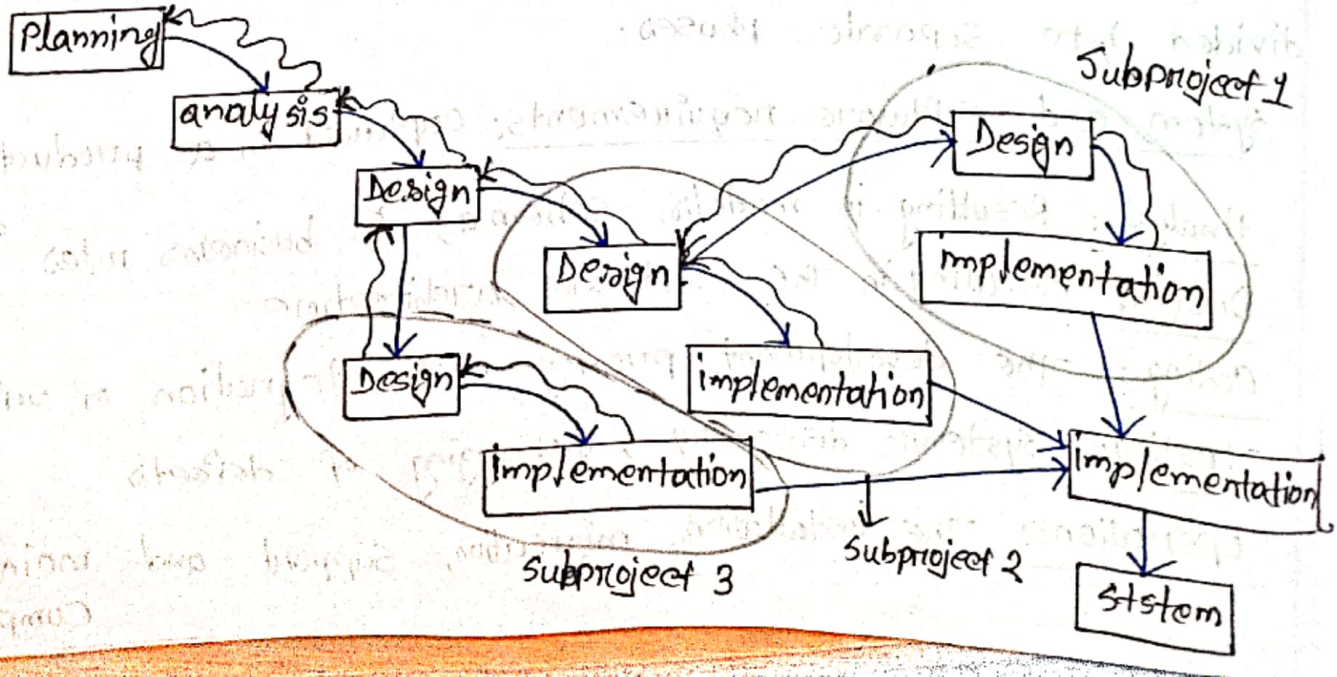


Waterfall development model

Variants of waterfall development: Two important variants.

① parallel development ② V-model

Parallel development:



⇒ It reduces the time required to deliver a system.

⇒ The project is divided into a series of subprojects that can be designed and implemented in parallel. Once all subprojects are complete there is a final integration.

⇒ If the subprojects are not completely independent, at the end of the project end, subprojects maybe quite challenging.

V-Model:

V-model means Verification and validation model. V model is another variation of waterfall development that pays more explicit attention to testing.

The development process proceeds down the left-hand slope of the V.

On the upward sloping right side of the model, unit testing, integration testing, system testing and finaly acceptance testing are performed.

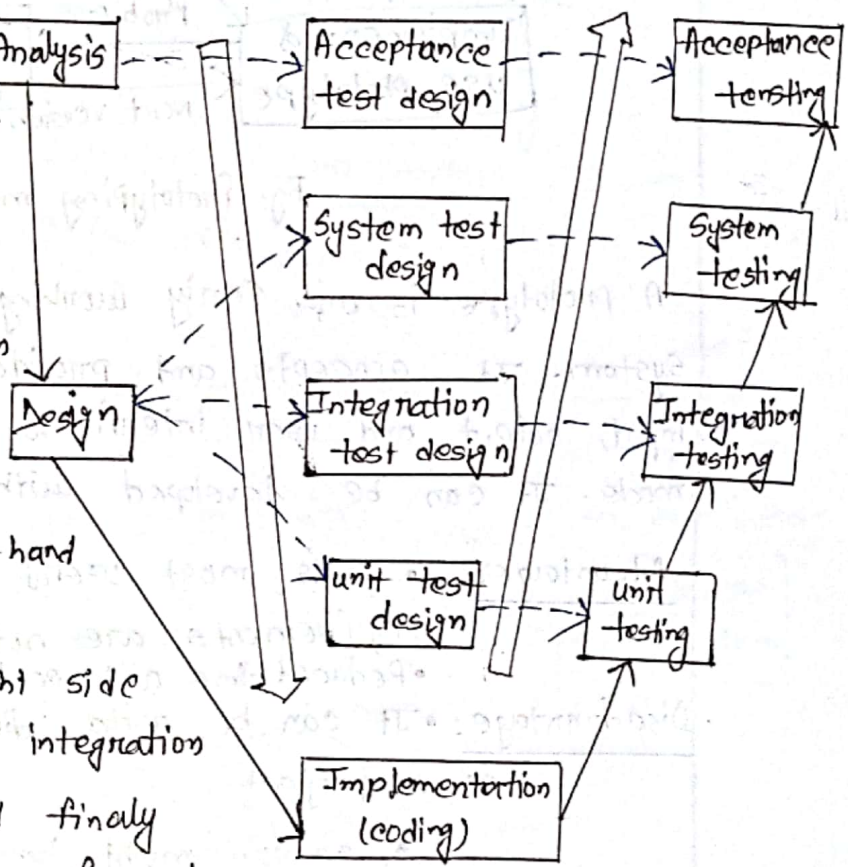


Figure: V-model

waterfall model advantage:

⇒ Simple and easy to understand
⇒ Clearly defined stages
⇒ Easy to arrange tasks
⇒ well understood milestones.

waterfall model disadvantage:

⇒ High amount of risk and uncertainty
⇒ It is difficult to measure progress within stages.
⇒ Cannot accommodate changing requirements

Prototyping:

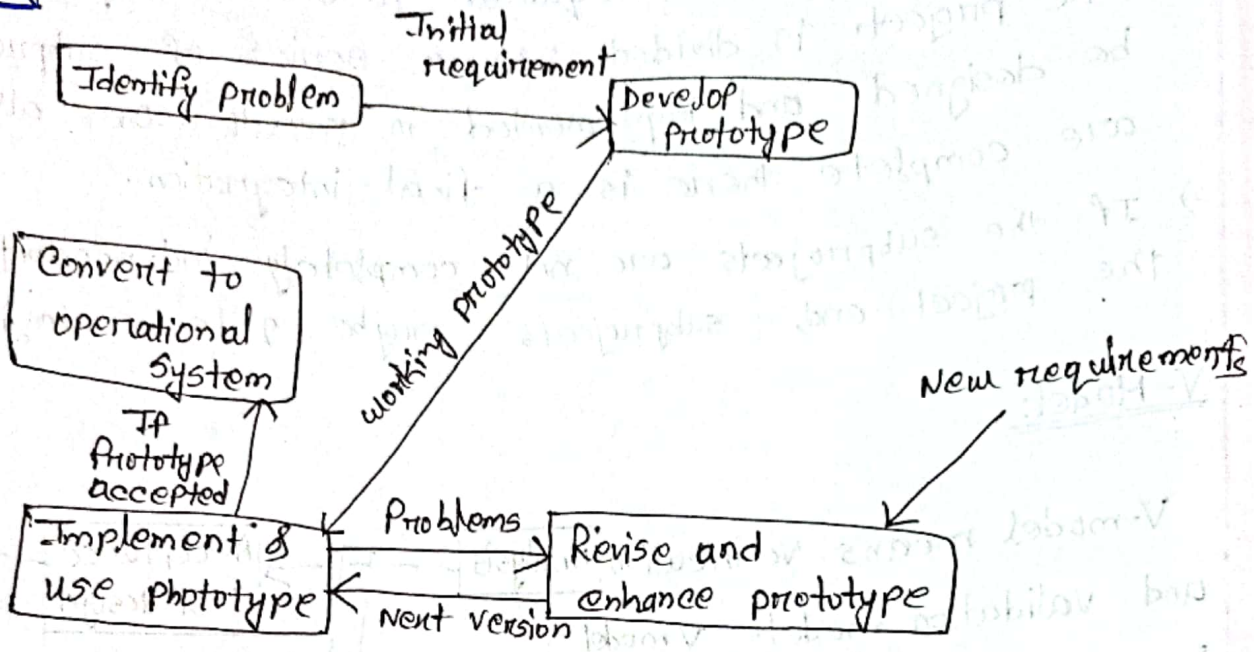


Fig: Prototyping method

A prototype is an early working version of an information system. It concepts and provides an opportunity to examine input, output and user interfaces before final decision are made. It can be developed with a CASE tool.

Advantages:
- It is most useful for requirements when user requirements are not clear.
- Reduced time and cost.

Disadvantage:
- It can be quite difficult to manage prototyping as a project.
- ⇒ Decision might be made too early
- User may get confused in the prototypes and actual systems.

application

RAD (Rapid development): Rapid is an object oriented approach to systems development that includes various software tools. CASE, JAD, visual programming languages, code generators may all play a role in RAD. RAD is a time based technique that speeds up information. RAD is the new information system.

RAD is an adaptive software development approach where a software prototype is rapidly updated based on user feedback and iteratively delivered until it meets all client requirements. 4 phases – 1. Requirements planning
2. Design system
3. Build the system
4. Implementation

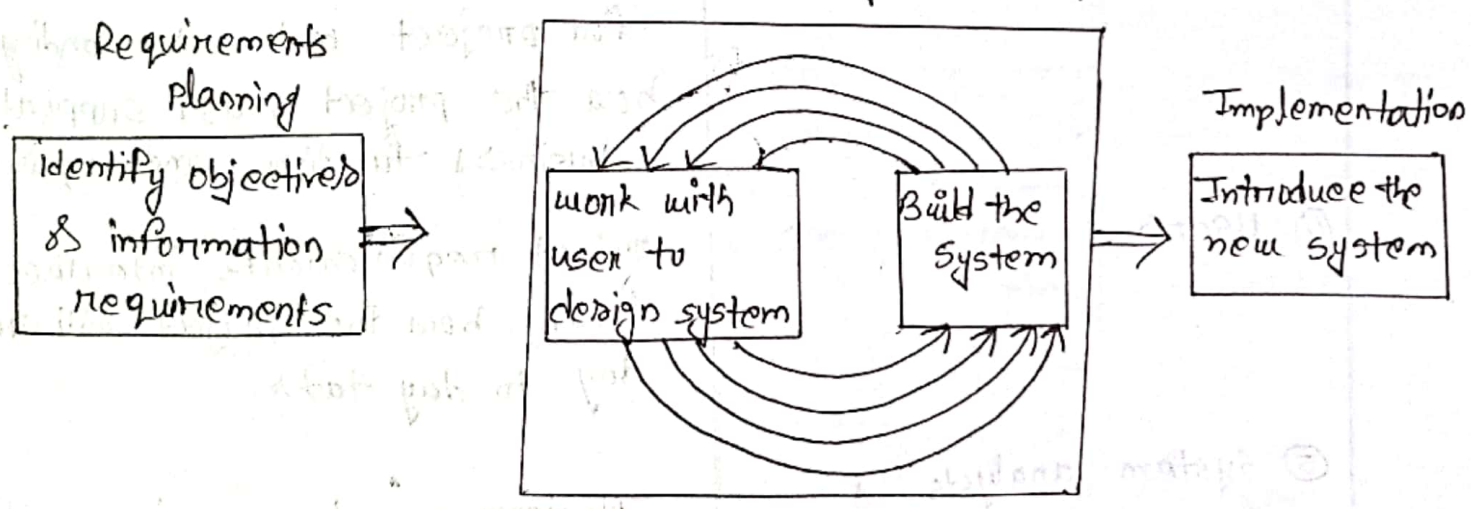


Fig: Four phases of the RAD model

Advantages:

⇒ Reduced development time
⇒ Increases reusability of components.
⇒ Encourages customer feedback.

Disadvantages:

⇒ Requires highly skilled developers.
⇒ High dependency on

Joint Application design (JAD): JAD is a structured process in which users, managers and analysts work together for several ~~work~~ days in a series of intensive meetings to specify or review system requirements. It was developed by IBM in the late 1970s.

JAD participant	Role
① JAD project leader	Develops an agend, acts as a facilator and leads the JAD session.
② Top management	Provides enterprise level and support for the project.
③ Managers	Provide department level support for project and understanding how the project must support business functions and requirements
④ Users	output requirements, interface issues, how the project will support day to day tasks.
⑤ System analysts & other IT staff members	JAD team members on issues such as security, backup, hardware, software and network capibility.
⑥ Recorder	Documents result of JAD sessions and works with systems to build system models and develop case tool documentation.

CASE tools: (computer aided software engineering)

It is software tools that assist systems builders in managing the complexities of information system. And help ensure that high quality systems are constructed on time and within budget. CASE tools are set of software application program. which are used to automate SDLC activities. CASE tools are used by software project managers, analysts and engineers to develop software system.

Example: Visible Analyst (software) is one example of a CASE tool that helps to do graphical planning.

General types of case tools:
- It enable system process
- It enables the storge of specification, diagram etc.
- Code generators that enable the automatic generation of program.

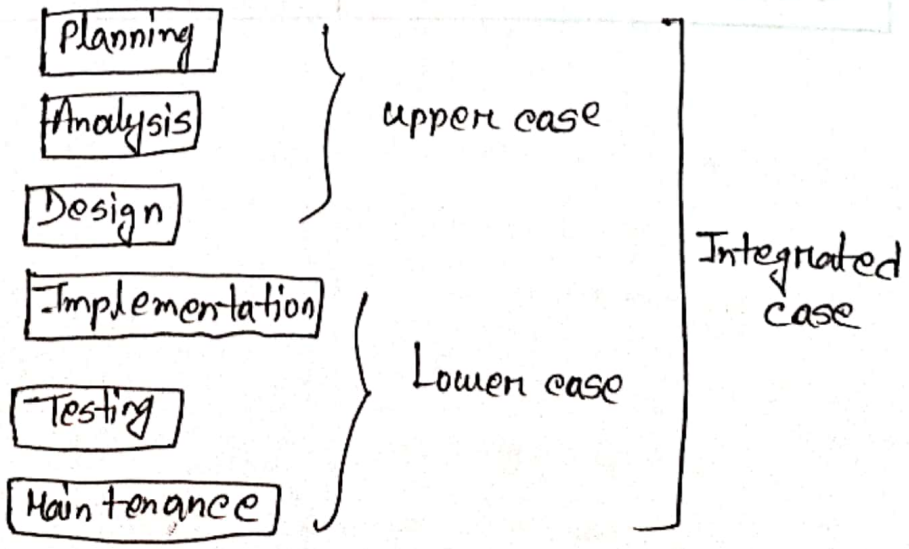
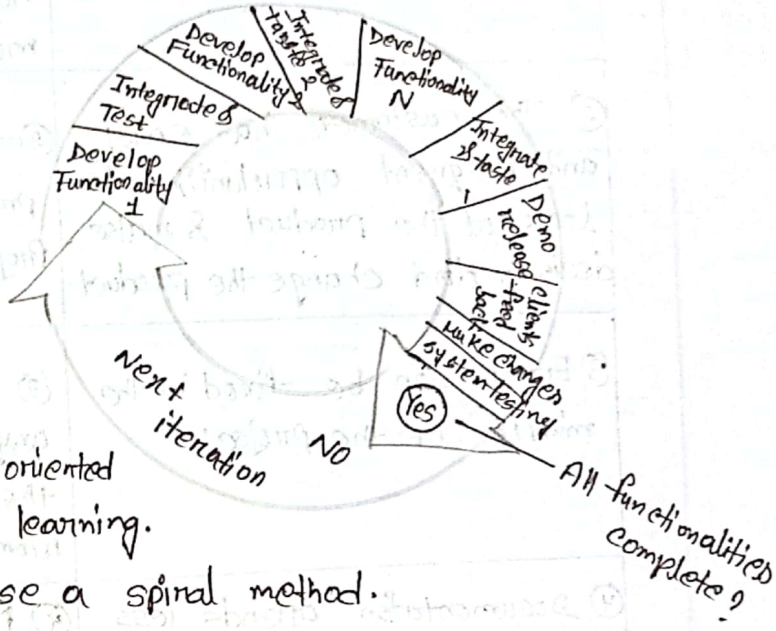


Fig: Case tools.

Agile method:

Agile method has evolved in the mid-1990s as a part of reaction against waterfall method.



⇒ Agile method is software development method that is people-focused communications-oriented flexible, speedy, responsive & learning.

⇒ Agile method typically use a spiral method.

⇒ There are many methodologies under agile method:
Adaptive software development, Scrum, RAD, extreme programming.

Advantages:

⇒ Adaptive method is very flexible.

⇒ Efficient in dealing with change.

⇒ Frequent deliverables constantly validate the project and reduce risk.

Disadvantage:

⇒ Team members need a high level of technical & interpersonal skill.

⇒ Lack of structured documentation
Can introduce risk factors.

⇒ Difficult to predict

⇒ Documentation is less.

⇒ Agile system analysis focuses on flexibility, Continuous improvement, managing uncertainty & speed.

Agile model Vs Waterfall model

Agile model	Waterfall model
① Agile model process is broken into individual models.	① The design process is not broken into an individuals models.
② The customer has early and frequent opportunities to look at the product & make decision and change the product.	② The customer can see only the product at the end of the Project
③ Error can be fixed in the middle of the project.	③ If Error is found on any changes have to be made, the project has to start from the beginning,
④ Documentation attends less priority the SD.	④ Documentation is a top priority.
⑤ Testers and developers work together.	⑤ Testers work separately from developers.
⑥ Agile is more flexible.	⑥ Waterfall is not as flexible as agile.
⑦ Every iteration has its own testing phase.	⑦ After the development phase, the testing phase is executed.

Spiral Model:

The spiral model is a combination of iterative development process model and sequential linear development model.

The life cycle is shown as a spiral starting in the center and working its way outward over & over again until the project is complete.

Each iteration produces feedback and enhancements, which enable the team to reach the overall project goal.

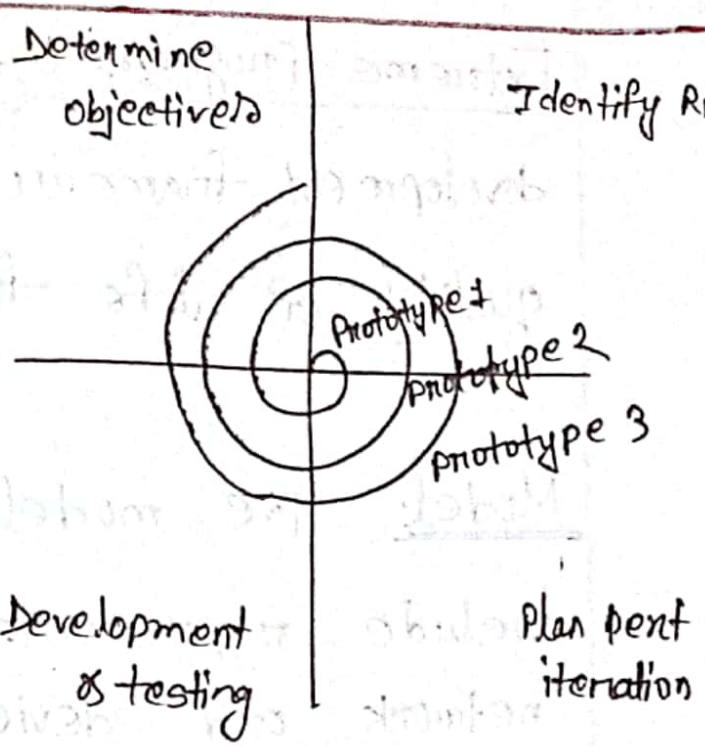


Fig: Spiral models

Model: The models used in system development include representations of inputs, processes, data, object, network and devices among other things.

A business model or requirement model; describes the information that a system must provide.

(a) Some model of system components

⇒ Flowchart

⇒ Data flow diagram (DFD)

⇒ Entity - relationship diagram (ERD)

⇒ use case diagram

⇒ class diagram

(b) Some models used to manage the development process

⇒ Gantt chart

⇒ Organizational hierarchy chart

⇒ Financial analysis models.

Tools: A tool is a software support system developers that helps create models or other components required in the project.

⇒ Types of tools:
- Drawing
- Visual modeling tool
- Code generator tool
- Database management ~~too~~ application

Technique: The technique indicate how the project team will do its work.

⇒ Project management techniques
⇒ Relational database design techniques
⇒ Structured programming technique
⇒ User interface design techniques

Approaches methodologies: ① Structured Approach
② ~~Structured analysis~~ Object oriented approach

Structured analysis: It helps the developer define what the system needs to do, what data system needs to store and use. The key graphical model of the system requirements that are used with structure analysis is called the data flow diagram (DFD).

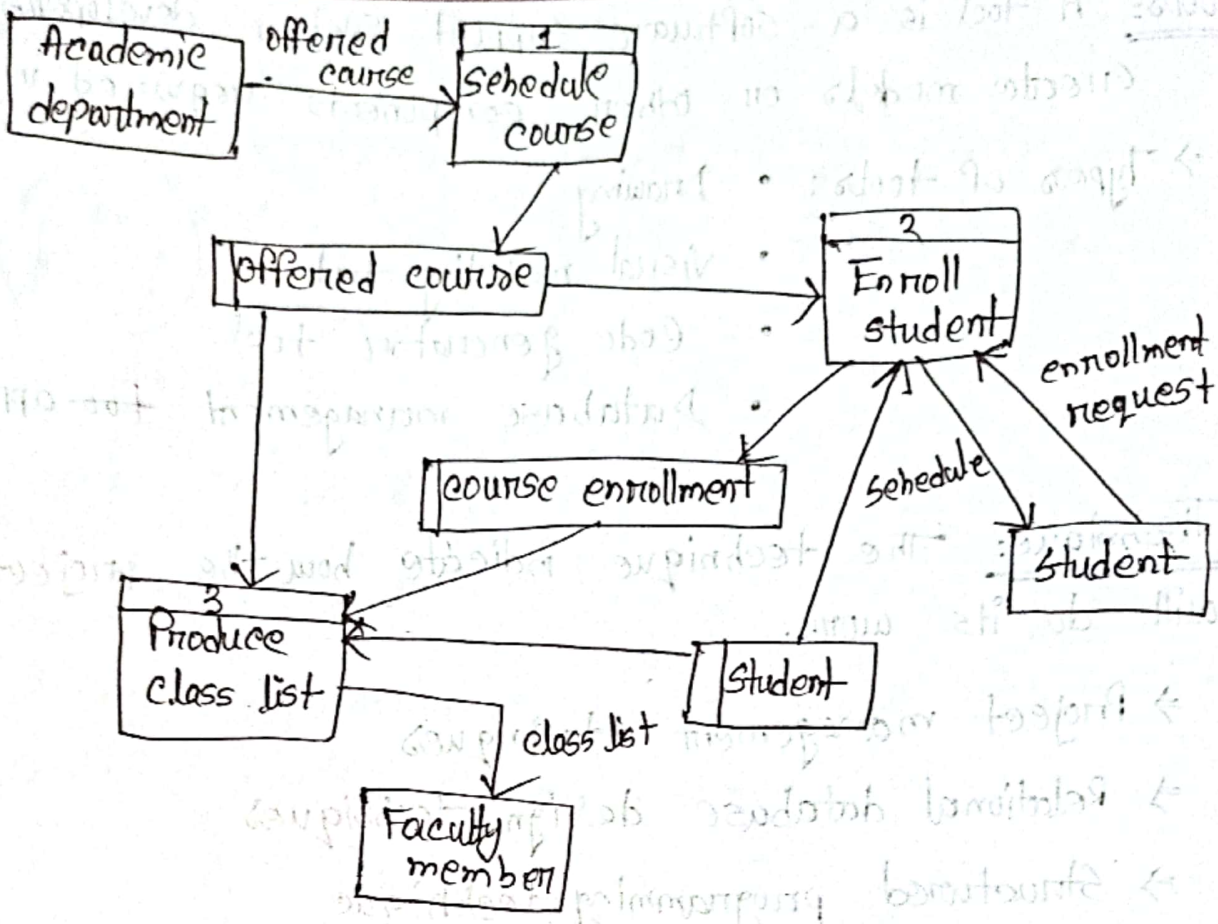


Figure: Data-flow diagram (DFD)

object oriented approach: The object oriented approach views an information system as a collection of interacting objects that work together to accomplish tasks.

⇒ There are no processes or programs, there are no data entities or files.

⇒ Code is modular and reuseable which can reduce cost and development time

⇒ More recent object-oriented languages include C++, Java and C#.

Increment development: Increment model are broken down into multiple standalone modules of sd. cycle. Increment development is always based on an iterative life cycle.

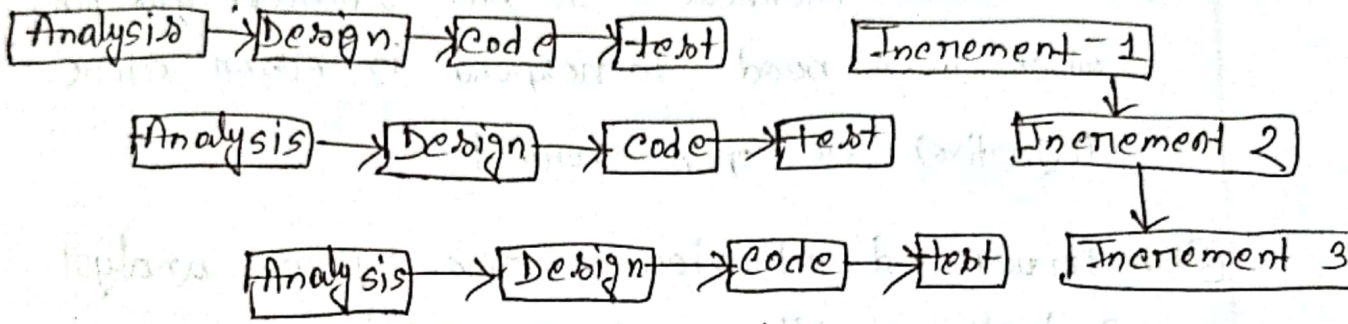


Fig: Increment model

Iterative Development: Iteration refers to the cycle nature of a process in which activities are repeated in a structure manner.

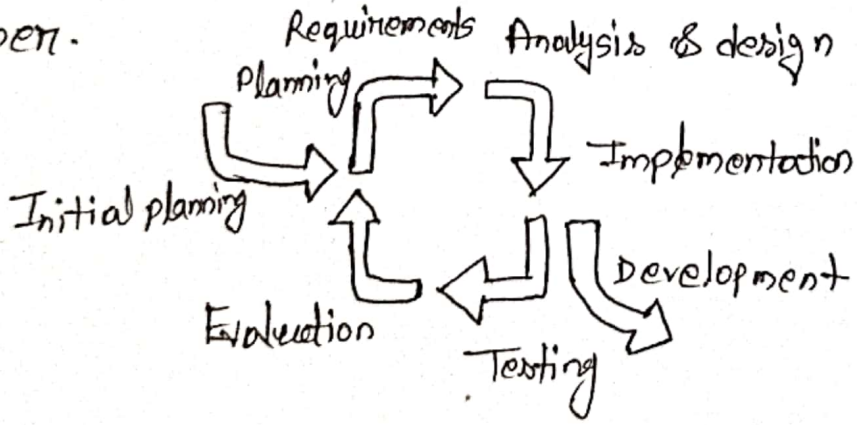


Fig: Typical iterative development process